MGMTFT/MGMTFE 410: Operations & Technology Management
Practice Final Examination — Exam 1

Professors Fernanda Bravo and Charles Corbett

Expectations of ethical conduct

As a member of the UCLA Anderson community, I am guided in my academic pursuits by the principles of integrity, respect, and honesty. I strive to embody these principles, not because I must, but because I believe it is the right thing to do.

For this exam, it is a violation of the honor code to receive or provide assistance by any means. Do not communicate with other students about the exam during the exam period. Do not use generative AI tools (ChatGPT, Claude, etc.) during the exam.

By signing this document, I acknowledge my obligations under the UCLA Anderson Honor Code and certify that I will not give or receive assistance during this examination.

_____ _____

Student's Signature (or name) Date

_____

Print Name

_____ _____

Student ID Section

Instructions

1. Print your name and student UID on the cover page.
2. You may ONLY use your cheatsheet and a calculator. You MAY NOT use a phone calculator, laptop, or any other device. You may not use notes other than your cheatsheet.
3. **Various definitions and formulas that we covered in class are provided at the end of this exam.**
4. Write your answers in the spaces provided.
5. For the open questions, in order to receive partial credit, you must show all your work.
6. For the multiple-choice questions, you do not need to show your work, and no partial credit will be awarded.
7. State any assumptions that you make.

Good Luck!!!

Question 1: miscellaneous multiple-choice questions. [16 POINTS]

1. [2 points] In a balanced serial process with five steps and one worker per step, which change will increase process capacity WITHOUT affecting throughput time?
   a. Adding a worker to the bottleneck step only.
   b. Adding a waiting buffer between the first and second steps.
   c. Reducing the processing time of the fastest (non-bottleneck) step.
   d. Adding a second worker to every step simultaneously.

   Answer: _____

2. [2 points] An outpatient clinic has an average of 45 patients present at any one time. On average, patients spend 3 hours at the clinic. By Little's Law, what is the flow rate of patients?
   a. 15 patients per hour.
   b. 135 patients per hour.
   c. 0.07 patients per hour.
   d. Cannot be determined from the information given.

   Answer: _____

3. [2 points] A project has five activities. P (3 wks) must precede R (4 wks). Q (4 wks) must precede S (5 wks). Both R and S must precede T (3 wks). P and Q can start simultaneously. What is the minimum project completion time?
   a. 15 weeks.
   b. 12 weeks.
   c. 9 weeks.
   d. 17 weeks.

   Answer: _____

4. [2 points] In the EOQ model, if the unit holding cost doubles (from h to 2h) while all other parameters stay the same, the optimal order quantity Q* will:
   a. Double.
   b. Decrease by approximately 29% (multiplied by $1/\sqrt{2}$).
   c. Stay the same.
   d. Quadruple.

   Answer: _____

5. [2 points] Daily demand ~ Normal(mean = 50, std dev = 20). Lead time = 4 days (constant). For a 95% service level (z = 1.645), what is the required safety stock?
   a. 16.4 units.
   b. 32.8 units.
   c. 65.6 units.
   d. 82.0 units.

   Answer: _____

6. [2 points] A food truck operator buys sandwiches for $8 each and sells them for $14 each. Unsold sandwiches have zero salvage value. What are C_u, C_o, and the critical fractile?
   a. C_u = $6, C_o = $8, critical fractile = 0.43.

b. C_u = \$8, C_o = \$8, critical fractile = 0.50.
c. C_u = \$6, C_o = \$6, critical fractile = 0.57.
d. C_u = \$14, C_o = \$8, critical fractile = 0.64.

Answer: _____

7. [2 points] A hospital has two independent triage stations each serving half of all patients. A consultant recommends pooling them into one shared queue with two servers. Compared to the current setup, pooling will result in:
   a. The same utilization and the same average waiting time.
   b. The same utilization but shorter average waiting time.
   c. Lower utilization and shorter average waiting time.
   d. Higher utilization and longer waiting time due to a single shared queue.

   Answer: _____

8. [2 points] What are some key benefits of reducing batch sizes in the Toyota Production System?
   a. Increases machine utilization by reducing idle time between batches.
   b. Increases setup time per unit, thereby justifying automation investment.
   c. Surfaces quality problems faster, reduces WIP inventory, and shortens throughput time.
   d. Reduces the need for cross-trained workers by specializing each worker on a single batch.

   Answer: _____

Question 2 — GreenWave Coffee (Queueing) [9 POINTS]

GreenWave Coffee has one barista station. Customers arrive at 20/hr (CV_a = 1.0). Average service time = 2 min (CV_s = 1.0).

a) [3 pts] What is the utilization of the barista station? Show your work.

b) [3 pts] GreenWave is considering adding a second barista who would share the same queue. Qualitatively (no calculation required): what will happen to (i) utilization, (ii) average wait time? Justify briefly.

c) [3 pts] A customer arrives and finds 3 customers ahead; the first of those 3 is just starting service. Service time = 2 min/customer (deterministic). How long does this arriving customer expect to wait before starting service?

Question 3 — MedKit Assembly (Process Analysis) [18 POINTS]

MedKit Assembly — six-step serial process (A–F). Assume no variability; each worker is dedicated to one step.

Step	A	B	C	D	E	F
Task time (min/unit)	15	40	20	50	25	30
Number of workers	1	2	1	3	1	2

For questions 3.1–3.5, circle the correct answer. For 3.6, show your work.

3.1. [2 pts] What is the minimum possible throughput time for the MedKit process?

e. 50 minutes
f. 115 minutes
g. 143 minutes
h. 180 minutes
i. None of the above

Answer: _____

3.2. [2 pts] What is the steady-state process capacity for the MedKit process?

j. 1.0 u/hr
k. 1.2 u/hr
l. 2.0 u/hr
m. 3.0 u/hr
n. None of the above

Answer: _____

3.3. [2 pts] Which step is the bottleneck in the MedKit process?

o. Step B
p. Step D
q. Step E
r. Step F
s. Steps B and D are tied

Answer: _____

3.4. [3 pts] One worker is moved from Step B to Step E (so B has 1 worker, E has 2). What is the new process capacity?

t. 1.2 u/hr
u. 1.5 u/hr
v. 2.0 u/hr
w. 2.4 u/hr
x. None of the above

Answer: _____

3.5. [3 pts] Returning to the original config: if all workers become fully flexible (same processing times, any worker can do any step), what is the minimum number of workers needed to achieve a capacity of at least 3 u/hr?

y. 3 workers
z. 4 workers
aa. 5 workers
bb. 6 workers
cc. 8 workers

Answer: _____

3.6. [6 pts] Redesign into 4 steps: W = 10, X = 60, Y = 30, Z = 20 min/unit with all 10 original workers. (a) Describe how you would allocate the 10 workers to maximize capacity, and (b) calculate the resulting capacity (u/hr).

Question 4 — SkyBridge Construction (Project Management) [15 POINTS]

SkyBridge Construction (HorizonBuild Inc.) — activity durations in weeks:

Activity	Description	Duration (wks)	Predecessors
A	Site preparation	3	—
B	Foundation excavation	4	A
C	Foundation pouring	6	B
D	Steel fabrication (off-site)	8	A
E	Steel delivery & inspection	2	D
F	Structural assembly	5	C, E
G	Decking installation	3	F
H	Safety railings & finishing	4	G
I	Final inspection & handover	2	H

a) [4 pts] Identify all paths through the network, list their durations, and identify the critical path. Show your work.

b) [4 pts] Shorten the project by 2 weeks at minimum cost. The following activities can be accelerated: D: $5,000/wk (max 2 wks) C: $6,000/wk (max 2 wks) F: $9,000/wk (max 1 wk) What is the minimum-cost acceleration plan and its total cost?

c) [3 pts] If activity durations are uncertain and independent, will the expected project duration be greater than, equal to, or less than 27 weeks? Justify your answer.

Question 5 — FlashFit Activewear (Inventory) [21 POINTS]

FlashFit Activewear — seasonal jacket (order once, no reorder).

Parameter	Value
Purchase cost	$60/unit
Full selling price	$150/unit
Outlet (salvage) price	$30/unit
Demand distribution	Normal, mean = 800, std dev = 200 units

Standard Normal CDF — use smallest z where P(Z ≤ z) ≥ CF:

z	P(Z ≤ z)	z	P(Z ≤ z)
0.0	0.500	1.1	0.864
0.1	0.540	1.2	0.885
0.2	0.579	1.3	0.903
0.3	0.618	1.4	0.919
0.4	0.655	1.5	0.933
0.5	0.691	1.6	0.945
0.6	0.726	1.7	0.955
0.7	0.758	1.8	0.964
0.8	0.788	1.9	0.971
0.9	0.816	2.0	0.977
1.0	0.841		

a) [3 pts] What are FlashFit's C_o and C_u? Explain each.

b) [4 pts] Find the critical fractile and optimal Q*. Show your work using the z-table above.

c) [3 pts] With Q* = 940, what is the probability of having unsold inventory at end of season?

d) [4 pts] A competitor offers to buy surplus jackets at end of season for $50 each instead of $30. Recalculate Q*. How and why does it change? Use the z-table above.

e) [3 pts] FlashFit considers moving to a domestic supplier allowing a mid-season reorder. How does this change the initial order quantity and expected profit? No calculation needed.

f) [4 pts] FlashFit also stocks T-shirts: annual demand 12,000 units, fixed order cost $200, annual holding cost $5/unit. (i) EOQ? (ii) Orders per year?

Question 6 — Miscellaneous Open-Text Questions [21 POINTS]

Each part is worth 3 points.

a) [3 pts] If managing your personal finances is your bottleneck, cite three strategies to better manage your finances and link each strategy explicitly to an operations management concept.

b) [3 pts] A fast-food chain is considering converting its three drive-through lanes (dedicated cashiers) into one shared lane (pooled cashier). Discuss the operational and behavioral considerations — both for and against pooling.

c) [3 pts] A grocery retailer orders its best-selling cereal twice a year in large bulk shipments. An analyst suggests ordering smaller quantities more frequently. What are the tradeoffs? Under what conditions would the analyst's suggestion be correct?

d) [3 pts] A ski resort must decide before the season how many daily lift passes to produce (printed in advance). Unsold passes have zero value; turning away a customer means losing the full ticket revenue. Using the newsvendor framework, explain what determines the optimal number of passes.

e) [3 pts] Toyota's TPS emphasizes making defects visible immediately rather than discovering them at the end of the line. What operational principle underlies this approach, and what are two practical benefits?

f) [3 pts] Describe two ways in which operations management concepts apply to non-manufacturing settings (service, knowledge work, or personal life). Be specific about which concepts you are applying.

Queueing theory formulas

Here are the queueing formulas, first the general approximations, and in the box to the right the special cases with m=1 server and with coefficients of variation of interarrival times and service times $CV_{IAT} = CV_{ST} = 1$.
Notation:

- A is the arrival rate (in customers per unit time)
- S is the service rate (in customers per unit time)
- m is the number of servers
- u is the utilization, defined as u = A/(m x S)
- CV_{IAT} : coefficient of variation of interarrival times = standard deviation / mean of interarrival times
- CV_{ST} : coefficient of variation of service times = standard deviation / mean of service times

- Average number in line $$L_q \approx \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2}$$
- Average time waiting in line $$W_q \approx \frac{1}{A} \times \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2}$$
- Average time in system $$W \approx \frac{1}{A} \times \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2} + \frac{1}{S}$$
- Average number in system $$L \approx \frac{u^{\sqrt{2(m+1)}}}{(1-u)} \times \frac{(CV_{IAT})^2 + (CV_{ST})^2}{2} + \frac{A}{S}$$

$$L_q = \frac{u^2}{1-u}$$

$$W_q = \frac{1}{S} \times \frac{u}{1-u}$$

$$W = \frac{1}{S} \times \frac{1}{1-u}$$

$$L = \frac{u}{1-u}$$

Inventory theory formulas

Here are the formulas we introduced for the various inventory models.
Economic Order Quantity (EOQ) model
Definitions:

- D = yearly demand rate (units/year)
- h = unit inventory holding costs ($/unit/year)
- K = fixed ordering cost (or set up cost) ($/order)
- Q = order quantity (units)

Optimal order quantity is the Economic Order Quantity:

- $Q^* = \sqrt{\frac{2\,K\,D}{h}}$

Newsvendor model
Definitions:

- D = (Random) Demand of mean μ and std dev σ.
- Probability distribution (e.g., Normal): $prob(D \leq Q)$
- Co = Overage Cost
- Cu = Underage Cost
- Q = Order Quantity
- critical fractile = cu/(co+cu)
- for given order quantity Q, the service level = $prob(D \leq Q)$

The newsvendor model prescribes that you should choose an optimal order quantity Q* such that the service level = the critical fractile, i.e., $prob(D \leq Q)$ = cu/(co+cu).
What is the service level (SL) if we know:

- the quantity Q? SL = NORM.DIST(Q, mean, stdev, TRUE)
- the z-score? SL = NORM.S.DIST(z, TRUE)

Given a service level (SL):

- what is the quantity Q? Q(SL) = NORM.INV(SL, mean, stdev)
- what is the z-score? z(SL) = NORM.S.INV(SL)

Need to find optimal Q^$*$ such that the service level satisfies:
$prob(D \leq Q$^$*)$ = NORM.DIST (Q^$*$, m, s, TRUE) = $cu/(cu+co$)
Use the inverse of the normal distribution:
$Q*$ = NORM.INV($cu/(cu+co$), m, s, TRUE)
Or use the inverse of the standard normal distribution:
z$*$= NORM.S.INV($cu/(cu+co$),TRUE)
Q^$*$ = μ + $\sigma \times z*$

Reorder point model
Definitions:

- D = (Random) Demand of mean μ and std dev σ. (units/period)
- Probability distribution (e.g., Normal): $prob(D \leq Q)$
- h = unit inventory holding costs ($/unit/period)
- K = fixed ordering cost (or set up cost) ($/order)
- LT = lead time (between placing an order and receiving the order) (periods)
- Q = Order Quantity
- target service level SL; for given order quantity Q, the service level = $prob(D \leq Q)$
- the z-score corresponding to service level SL = z_{SL}

The reorder point model has two parameters: the order quantity Q, and the reorder point ROP.
For the order quantity Q, use the EOQ formula.
For the reorder point: ROP = $\mu \times LT + z_{SL} \times \sigma_D \times \sqrt{LT}$

Process analysis definitions

- Cycle Time: Time between completion of successive units (e.g., minutes / patient, hours / customer)
- Capacity: How much can be processed per unit time (e.g., patients / minute, customers / hour)
- Flow Rate: How much is actually processed per unit time
    - Flow rate = minimum{Demand, Process Capacity}
- Utilization: proportion (or %) of capacity used
    - Utilization = Flow Rate / Capacity , OR
    - Utilization = Time Occupied / Time Available
- Bottleneck: Resource which limits the flow
    - Often (in simple processes): bottleneck is step with the lowest capacity, or step with highest utilization
- Balance/Imbalance: Balanced process if all stages have equal capacity
- Idle Time (IT): Time spent per cycle not doing useful work
- Direct Labor Content (DLC): Total operator time spent per unit of product (i.e., per cycle)
- Direct Labor Utilization (DLU): Percentage of operator time spent doing useful work
    - DLU = DLC / (DLC + IT)
    - DLU = DLC / (total operator time per cycle)
- Throughput Time (TT): Sum of processing and wait times at all stages of the process
- Lot (Batch) Size: Number of units of the same product produced together before starting a new batch
- Setup Time: Preparation time before batch is produced
- Run Time: Actual time to produce the batch, excluding the setup time